

01 Linear-Truth Cross-Validation Replication

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Goal

This project simulates data under a linear truth and compares two candidate models with one random 10-fold split per Monte Carlo replication:

- Linear model (correct): $y \sim x$
- Polynomial model: $y \sim \text{poly}(x, 3)$

The aim is to show that CV model selection still varies across finite replications.

Simulation settings

Table 1: Simulation settings.

setting	value
K	10
Monte Carlo replications per sample size	1000
Sample sizes	10000, 30000
beta0	1
beta1	2
sigma	4
poly_degree	3

Selection Counts and Rates

Table 2: How often each model is selected across 1000 replications for each sample size.

n	selected_model	count	selection_rate
10000	linear	841	0.841
10000	poly3	159	0.159
30000	linear	853	0.853
30000	poly3	147	0.147

CV Error Summary

Table 3: Average CV error and gap summary by sample size (poly3 minus linear).

n	mean_cv_linear	mean_cv_poly3	mean_gap_poly_minus_linear	sd_gap_poly_minus_linear
10000	16.01091	16.01436	0.00345	0.00414
30000	15.99943	16.00061	0.00119	0.00125

Conclusion

Even when the true model is linear, cross-validation can select either linear or polynomial in finite replications. The selection-count table quantifies this variation for both large-sample datasets ($n = 10000$ and $n = 30000$).